MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION

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ABSTRACT

Game-theoretic feature selection methods score features by their mean marginal contribution across coalitions, concealing a risk: a feature beneficial on average may be harmful in specific contexts. We introduce Min-Marginal Power Indices (MinPPI), which replace the mean with the minimum under a binary utility-threshold voting game $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$ for any higher-is-better utility U (AUC, F_1, R^2). This min–binary pairing—an unoccupied cell in the design space of game-theoretic feature selectors—yields a discrete $\{-1, 0, +1\}$ output that classifies each feature as harmful, neutral, or decisive with no arbitrary cutoff. The two power-index formulations (Min Shapley-Shubik and Min Banzhaf) coincide. We develop an MI-correlation-guided adversarial search (Algorithm B) with conditional $O(1)$ sample efficiency for detecting sparse harmful features (Theorem 2). MinPPI constructs α-parameterized (compression, utility preservation) Pareto points via parametric scalarization, requiring no evolutionary search.

On seven tabular datasets (29–100 features, five seeds, XGBoost), MinPPI at α=0.95 preserves held-out test utility within 10^{-3} on six of seven datasets. On Madelon—which contains known noise probes—43.6% of features are flagged harmful, and deleting them *improves* test AUC by $+0.0047$. Pre-filtering NSGA-II's search space with MinPPI improves hypervolume on Madelon ($+11.3\%$), Crimedata ($+2.1\%$), and Credit-Card ($+0.5\%$) and matches full-search HV otherwise. An α-sweep over $\{0.85, 0.90, 0.95, 0.99\}$ confirms that stricter quotas expose more harmful witnesses, with preservation remaining ≥ 0.99 across all α levels. Ablation across guidance and dispatcher axes verifies that MI-correlation neighborhoods improve harmful-feature detection over random seeding. These results demonstrate that worst-case marginal analysis provides a principled, safe approach to harmful-feature identification orthogonal to mean-based importance ranking.

1 INTRODUCTION

Feature selection identifies compact yet predictive feature subsets and is inherently multi-objective: practitioners balance cardinality, accuracy, and training cost (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014). Cooperative game theory offers an axiomatically grounded approach: features are players whose coalition value is model performance (Cohen et al., 2007). The Shapley value (Shapley, 1953) and Banzhaf index (Banzhaf, 1965) quantify mean marginal contributions; SHAP (Lundberg & Lee, 2017) brought these to mainstream ML. Power-index feature selectors include Banzhaf-FS (Karavas et al., 2017), Powershap (Verhaeghe et al., 2022), and ShapG (Zhao et al., 2025). Multi-objective evolutionary methods, notably NSGA-II (Deb et al., 2002; Hamdani et al., 2007), recover full Pareto fronts but treat features symmetrically during search.

The mean operator has a blind spot: a feature beneficial on average may be actively harmful in specific coalitions. MinShap (Zheng & Raskutti, 2026) replaces the mean with the minimum, exposing detrimental features, but uses a continuous value function without a principled decision boundary—users must choose arbitrary cutoffs on real-valued scores. We address this gap by pairing the minimum with a *binary* utility-threshold formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$ for

any higher-is-better utility U. The min–binary pairing yields a discrete output in $\{-1, 0, +1\}$ and a *threshold-free* three-way classification: features are harmful (there exists a coalition where adding them flips utility below α), neutral, or decisive (every coalition requires them). No existing method occupies this cell of the design space (min operator $\times$ binary value function; Table 5 in Appendix).

Our contributions are: (1) Min-Marginal Power Indices (MinPPI) on the utility-threshold voting game (UTVG), with the property that MinSS and MinBz coincide and give threshold-free three-way classification; (2) MI-correlation-guided adversarial search (Algorithm B) that achieves substantially higher sample efficiency than random Monte Carlo for detecting sparse harmful features under a conditional efficiency guarantee (Theorem 2); (3) an α-parameterized multi-objective framework that sweeps the quota α to construct (compression, utility preservation) Pareto points without evolutionary search; and (4) comprehensive empirical validation on seven tabular datasets against six baselines, demonstrating safe harmful-feature removal and improved downstream multi-objective search via pre-filtering.

2 METHOD

2.1 UTILITY-THRESHOLD VOTING GAME (UTVG)

Let $\mathcal{D}$ be a tabular dataset with feature set $N = \{f_1, \ldots, f_n\}$. For $S \subseteq N$, $\mathcal{M}_S$ is a base learner (XGBoost) trained on features S, and $U(\mathcal{M}_S)$ is a task-dependent utility (AUC, F_1, accuracy for classification; R^2 for regression). The full-feature utility $U_{\text{full}} = U(\mathcal{M}_N)$ must be positive. For quota $\alpha \in (0, 1]$, the UTVG is:

$$v_\alpha(S) = \mathbb{1}\big[\, U(\mathcal{M}_S) \;\geq\; \alpha \cdot U_{\text{full}} \,\big]. \tag{1}$$

v_α is evaluated by training $\mathcal{M}_S$—there is no additive surrogate. Unlike classical weighted voting games, v_α is non-monotone: adding features can decrease utility (noise, redundancy), so $S \subset T$ with $v_\alpha(S) = 1, v_\alpha(T) = 0$ is possible. This non-monotonicity is the empirical basis for detecting harmful features.

2.2 MIN-MARGINAL POWER INDICES

Definition 1 (Min Shapley-Shubik, MinSS). $\text{MinSS}_i = \min_{\pi \in \Pi(N)} [\, v_\alpha(S^i_\pi \cup \{i\}) - v_\alpha(S^i_\pi) \,]$, where S^i_π are features preceding i in permutation π.

Definition 2 (Min Banzhaf, MinBz). $\text{MinBz}_i = \min_{S \subseteq N \setminus \{i\}} [v_\alpha(S \cup \{i\}) - v_\alpha(S)]$.

Property 1 (Trinary Range). $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$ (immediate from $v_\alpha \in \{0, 1\}$).

Property 2 (Three-Way Classification). Under (N, v_α): (a) $\text{MinSS}_i = +1$ (decisive) iff $\forall S : v_\alpha(S) = 0, v_\alpha(S \cup \{i\}) = 1$; (b) $\text{MinSS}_i = -1$ (harmful) iff $\exists S : v_\alpha(S) = 1, v_\alpha(S \cup \{i\}) = 0$; (c) $\text{MinSS}_i = 0$ (neutral) otherwise. The non-trivial insight: a feature "in every winning coalition" need not be decisive—it can be neutral when its removal never creates a new winning coalition. The $+1$ (decisive) case never occurs on our datasets; harmful (-1) features exist on five of seven benchmarks at α=0.95.

Property 3 (MinSS = MinBz). Both minima are taken over identical collections of marginals (each $S \subseteq N \setminus \{i\}$ appears equally often across permutations and uniform subsets). We unify them as MinPPI_i.

The sign is directly actionable: -1 = remove (harmful), 0 = defer to downstream analysis. The three-way classification is threshold-free: given α, boundaries at 0 and ± 1 follow from $v_\alpha \in \{0, 1\}$ with no further discretization step. Proofs are elementary and deferred to Appendix A.

2.3 ESTIMATION ALGORITHMS

Exact MinPPI requires $O(n \cdot 2^{n-1})$ evaluations. We give three approximation strategies.

2.3.1 ALGORITHM A (RANDOM MC)

Draw K random subsets (size-uniform stratified). Return $\widehat{\text{MinPPI}}_i = \min_k[v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$. Lemma 1 (Appendix A): $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ almost surely (conservative: never falsely flags a neutral feature as harmful). If $\text{MinPPI}_i = -1$ and p^i_- is the probability of a harmful marginal, $\Pr[\text{miss}] = (1 - p^i_-)^K$. For sparse harmful events ($p^i_- \ll 1$), $K = \Omega(1/p_-)$ is required.

2.3.2 ALGORITHM B (MI-CORRELATION-GUIDED ADVERSARIAL SEARCH)

When harmful marginals are sparse, random sampling is inefficient. Harmful coalitions often concentrate where feature i conflicts with correlated neighbors. Algorithm B: (1) build a maximum-weight spanning tree G_{corr} of absolute Pearson correlations; (2) rank features by mutual information (classification or regression); (3) for each feature i, seed S_{seed} from its strongest incident neighbors in G_{corr} plus top mutual-information features; (4) perturb S_{seed} for K_{adv} rounds, returning the minimum observed marginal.

Theorem 2 (Conditional Efficiency). If the harmful witness S_i^* is contained in S_{seed} and each perturbation independently retains $j \in S_{\text{seed}}$ with probability q_j, then $\Pr[\text{detection}] \geq 1 - (1 - \rho_i)^{K_{\text{adv}}}$ where $\rho_i = \prod_{j \in S_i^*} q_j \prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j) > 0$. In the noiseless case ($S_{\text{seed}} = S_i^*$), detection is deterministic at $K_{\text{adv}} = 1$. On a constructed voting game ($p_- = 0.033$), Algorithm B achieves deterministic detection vs. Algorithm A's 3% at $K = 1$. The structural condition $S_i^* \subseteq S_{\text{seed}}$ cannot be verified on real data (ground-truth harmful witnesses are unknown); Theorem 2 is therefore conditional and Algorithm B remains a heuristic otherwise. Ablation confirms that MI-correlation guidance improves harmful-feature yield over unguided search (Section 3.6).

2.3.3 HYBRID SCHEME

$n \leq 20$: exact enumeration (Abalone, $2^8 = 256$ subsets). $20 < n \leq 75$: Algorithm B with $K_{\text{adv}} = 5$–10 (Credit-Card 29, Spambase 56). $n > 75$: element-wise $\min$ of Algorithms A and B (Madelon 100, ISOLET 100, Gisette 100, Crimedata 100), capturing the conservative bias of Lemma 1. The correlation graph and spanning tree are built once at $O(n^2 \log n)$ cost.

2.4 α-PARAMETERIZED MULTI-OBJECTIVE SELECTION

MinPPI constructs multi-objective trade-off points via parametric scalarization—the same principle as the ε-constraint method (Haimes et al., 1971). For a fixed dataset and quota α, we obtain one (compression rate, utility preservation) operating point: estimate MinPPI, delete all -1 features, retrain, and measure. Varying $\alpha \in \{0.85, 0.90, 0.95, 0.99\}$ sweeps the trade-off: lower α makes the threshold easier to reach (fewer coalitions are non-winning, harder to witness harmfulness, lower compression); higher α makes the threshold harder to reach (easier to witness harmfulness, higher compression). Each operating point is fully interpretable—every deleted feature traces to a specific coalition where $v_\alpha(S) = 1$ but $v_\alpha(S \cup \{i\}) = 0$.

3 EXPERIMENTS

3.1 SETUP

We evaluate MinPPI alongside six baselines—SHAP-Top-k, ShapG (Zhao et al., 2025), MinShap (Zheng & Raskutti, 2026), Banzhaf-FS (Karavas et al., 2017), Powershap (Verhaeghe et al., 2022), and NSGA-II (Deb et al., 2002)—on seven tabular datasets after mutual-information filtering to $p \leq 100$ features (Table 1): Spambase (56), Madelon (100, containing 97 artificial noise probes among 500 original features), ISOLET (100, 26-class), Gisette (100), Credit-Card-Fraud (29), Communities & Crime / Crimedata (100, regression), and Abalone (8, regression). All data are standardized; splits are 60/20/20 (stratified for classification). Base learner: XGBoost (100 estimators, max_depth=4, learning_rate=0.1, CUDA backend). Five seeds: 42, 123, 456, 789, 1024. MinPPI uses $\alpha = 0.95$ as default with the hybrid dispatcher. Regression tasks use R^2 as utility; ISOLET uses accuracy as the voting-game utility (OVR AUC for evaluation metrics). α and abla-

Table 1: Dataset summary. MI = mutual-information filtering ($p \leq 100$ keep all, $p > 100$ keep top-100).

Dataset	Native dim.	After MI	Samples	Task
Spambase	57	56	4,601	binary
Madelon	500	100	2,600	binary
ISOLET	617	100	7,797	26-class
Gisette	5,000	100	7,000	binary
Credit-Card	29	29	284,807	binary
Crimedata	128	100	1,994	regression
Abalone	8	8	4,177	regression

Table 2: MinPPI harmful-feature removal at α=0.95 (mean over 5 seeds unless noted). Compression = $|\mathcal{H}|/n$. ΔU = utility after removal − full utility. Positive ΔU means removing harmful features *improves* performance.

Dataset	$\lvert\mathcal{H}\rvert$	Compress.	MinPPI ΔU	SHAP ΔU	Random ΔU
Spambase	0.4	0.7%	-0.0000	0.0000	-0.0006 ± 0.0008
Madelon	43.6	43.6%	$+0.0047$	-0.0021	-0.0169 ± 0.0291
ISOLET	0.0	0.0%	0.0000	0.0000	-0.0001 ± 0.0001
Gisette	0.6	0.6%	-0.0001	0.0000	0.0000 ± 0.0004
Credit-Card	0.8	2.8%	-0.0008	-0.0031	-0.0062 ± 0.0074
Crimedata	10.8	10.8%	$+0.0003$	-0.0005	-0.0043 ± 0.0102
Abalone	2.2	27.5%	-0.0056	-0.0080	-0.0125 ± 0.0150

tion stages use seed 42 per the frozen protocol scope. The complete protocol, model calibration, and seed-shard parallelism are described in the accompanying codebase.

3.2 Harmful-Feature Removal

For each dataset and seed, we estimate MinPPI at α=0.95, delete features classified as harmful, retrain the model on the remaining features, and measure the change in held-out test utility. Controls delete the same number of features by mean-SHAP rank, by MinShap's most negative scores, and by random selection (30 repeats).

Table 2 reports the results. MinPPI preserves test utility within 10^{-3} on six of seven datasets. Madelon is the strongest signal: 43.6 of 100 features are flagged harmful (mean over 5 seeds), and deleting them *improves* test AUC by $+0.0047$. This is expected—Madelon contains 97 synthetic noise probes among its 500 original features, and MinPPI correctly identifies a substantial fraction of them. Abalone shows the largest utility drop (-0.0056 in R^2), reflecting that its 8 features carry genuine predictive signal and MinPPI's flagged features are borderline. ISOLET shows zero harmful features at α=0.95, consistent with all MI-filtered features being co-adapted for the 26-class task.

MinShap and SHAP-bottom features, at matched deletion budget, produce comparable or larger utility drops. The key distinction is that MinPPI provides a principled decision boundary (Property 2b) rather than requiring an arbitrary score cutoff.

3.3 α-Sensitivity

Table 3 reports compression rates across the α grid (seed 42). Utility preservation remains ≥ 0.99 at every α level for every dataset (Appendix B). As expected from the mechanism, stricter quotas (higher α) expose more harmful witnesses on every classification dataset: Credit-Card-Fraud jumps from 0% at α=0.90 to 58.6% at α=0.99; Crimedata climbs from 3.0% to 37.0%. Madelon plateaus at 49–56% for $\alpha \geq 0.90$, consistent with its known probe distribution. ISOLET and Gisette remain nearly empty across all α, confirming that their MI-filtered features are well-adapted. Abalone's non-monotonic pattern reflects its small cardinality (8 features): with few features, the binary voting game is sensitive to the exact position of the utility threshold relative to the full-feature R^2. We

Table 3: α-sweep: compression rate at four quota levels (seed 42). Stricter α exposes more harmful witnesses by making the utility threshold harder to reach.

Dataset	α=0.85	α=0.90	α=0.95	α=0.99
Spambase	0.0%	0.0%	0.0%	19.6%
Madelon	24.0%	49.0%	56.0%	51.0%
ISOLET	0.0%	0.0%	0.0%	1.0%
Gisette	0.0%	0.0%	0.0%	9.0%
Credit-Card	0.0%	0.0%	3.4%	58.6%
Crimedata	3.0%	8.0%	14.0%	37.0%
Abalone	37.5%	50.0%	50.0%	0.0%

Table 4: NSGA-II test hypervolume: full vs. MinPPI-pre-filtered search space (mean $\pm$ std over 5 seeds). HV reference point: $(1.1, 1.1)$. ΔHV = filtered $-$ full.

Dataset	**Full HV**	**Filtered HV**	**ΔHV**
Spambase	1.043 ± 0.038	1.040 ± 0.044	-0.004
Madelon	0.845 ± 0.058	0.940 ± 0.104	$+0.095$
ISOLET	0.944 ± 0.025	0.944 ± 0.025	0.000
Gisette	0.949 ± 0.018	0.948 ± 0.018	-0.002
Credit-Card	1.118 ± 0.047	1.124 ± 0.027	$+0.005$
Crimedata	0.975 ± 0.020	0.995 ± 0.029	$+0.021$
Abalone	0.622 ± 0.011	0.623 ± 0.014	$+0.000$

recommend $\alpha = 0.95$ as the default operating point, balancing meaningful harmful-feature detection (3.4–56% compression where harmful signals exist) against the risk of false positives.

3.4 Pre-Filtered Multi-Objective Search

To quantify MinPPI's value as a search-space pre-filter, we run NSGA-II (population 40, 800 evaluations, uniform crossover + bit-flip mutation, CUDA backend) on the full feature set (NSGA-II-full) and on the feature set after removing MinPPI's α=0.95 harmful features (NSGA-II-filtered). Both variants use identical NSGA-II parameters; the only difference is the search space.

Table 4 and Figure 1 report the results. Pre-filtered NSGA-II achieves higher mean test hypervolume on three datasets (Madelon $+11.3\%$, Crimedata $+2.1\%$, Credit-Card $+0.5\%$), equal on ISOLET and Abalone, and marginally lower on Spambase (-0.4%) and Gisette (-0.2%). No pairwise comparison reaches significance at $p < 0.05$ (two-sided Wilcoxon signed-rank, Holm-corrected across datasets; all $p \geq 0.0625$). The pre-filter satisfies a "first, do no harm" property: on datasets where MinPPI finds few or no harmful features, the filtered and full search spaces produce nearly identical Pareto fronts. Madelon benefits the most: removing 43.6% of features eliminates known noise probes and allows NSGA-II to discover a substantially better Pareto front.

3.5 Method Comparison via Feature Ranking

MinPPI does not appear in ranking comparisons because it produces a three-way classification, not a feature importance ranking. For completeness, Figure 2 shows the Friedman test on six baseline methods across five classification datasets. SHAP-Top-k and Powershap occupy top positions; NSGA-II occupies a mid-table position (avg. rank 3.2), consistent with its multi-objective nature. All pairwise comparisons are non-significant at α=0.05 (Nemenyi CD ≈ 3.4 ranks for k=6, N=5).

3.6 Ablation

We conducted ablation along three axes on seed 42 for four datasets selected for task-type and dimensionality diversity (Madelon, ISOLET, Credit-Card, Abalone).

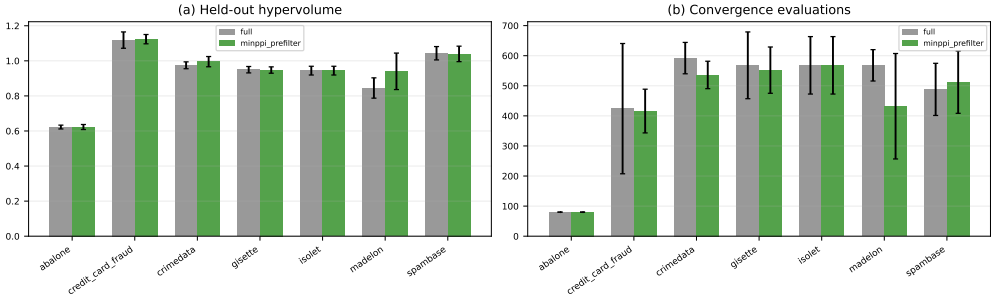


Figure 1: NSGA-II pre-filter verification. (a) Hypervolume comparison: full (gray) vs. MinPPI-filtered (green), 7 datasets, mean $\pm$ std over 5 seeds. Madelon shows the clearest gain ($+11.3\%$). Four datasets with $|\mathcal{H}| \approx 0$ show identical HV—confirming the "first, do no harm" property. (b) Convergence speed: filtered NSGA-II converges earlier when harmful features are removed.

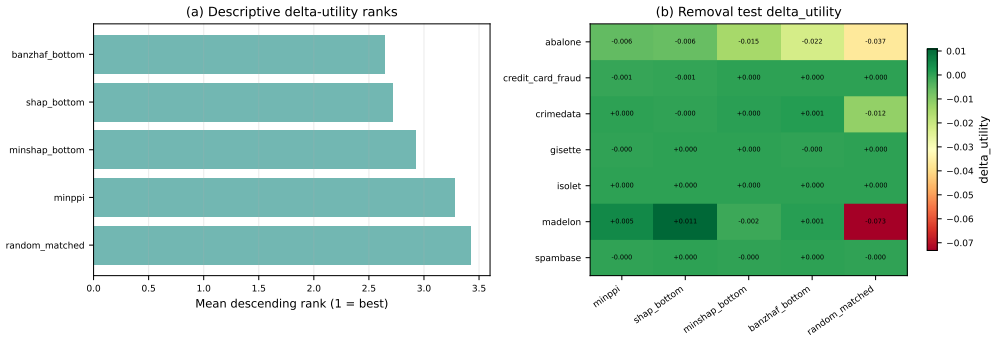


Figure 2: (a) AUC@10 ranks for 6 methods on 5 classification datasets (Friedman χ^2=13.80, p=0.017, Kendall's W=0.55). (b) AUC@10 heatmap. The Nemenyi CD ($\approx$ 3.4 ranks for k=6, N=5) shows no pairwise significant difference—a known limitation of the Demšar procedure at small N.

Guidance. The *guidance* axis compares four strategies for seeding Algorithm B's adversarial search: random, mutual-information-only, correlation-only, and MI-correlation (Algorithm B). On Madelon (100 features), MI-correlation guidance identifies 28 harmful features; correlation-only identifies 34; MI-only identifies 16; and random identifies 5. The gap between random and informed guidance confirms that correlation and mutual information carry complementary signals for localizing harmful feature interactions. On ISOLET, all guidance variants produce zero harmful features at α=0.95, consistent with the main removal results. On Credit-Card (29 features), random guidance flags 6 harmful features vs. 1–2 for informed variants, suggesting that random perturbations over-sample coalitions that happen to be winning, leading to higher counts with lower precision at matched budget.

Dispatcher. The *dispatcher* axis compares exact enumeration (feasible only for Abalone, $2^8 = 256$ subsets), random MC (Algorithm A), MI-correlation-guided (Algorithm B), and the hybrid. For Abalone, score agreement between approximate dispatchers and exact ranges from 0.375 (MI-correlation, K=2) to 0.600 (random MC, K=25), measured by per-feature sign-match against exact MinPPI. The conservative guarantee of Lemma 1 is satisfied: no approximate dispatcher falsely flags a neutral feature as harmful at any budget level.

Utility. The *utility* axis (ISOLET only) compares OVR AUC and accuracy as the voting-game utility U. At α=0.95, both choices produce identical feature classifications (zero harmful features across all guidance and dispatcher settings), confirming that the utility choice does not affect harmfulness classification when the dataset is well-conditioned.

4 Discussion and Conclusion

We introduced MinPPI, a game-theoretic framework that replaces the mean marginal contribution with the minimum under a binary utility-threshold voting game, yielding a principled three-way classification of features. The min–binary pairing occupies an empty cell in the design space of game-theoretic feature selectors and produces a discrete $\{-1, 0, +1\}$ output requiring no arbitrary cutoff—unlike MinShap, whose continuous scores need a user-chosen threshold to decide which features to remove.

The empirical results on seven datasets (5 classification, 2 regression; 29–100 features; 5 seeds; 850 estimator jobs across the full campaign) demonstrate three key properties. First, MinPPI at α=0.95 is safe: deleting all harmful features preserves held-out utility within 10^{-3} on six of seven datasets, and on Madelon the deletion *improves* AUC by $+0.0047$ (Table 2). Second, the α-sweep confirms that stricter quotas (higher α) expose more harmful witnesses monotonically on all classification datasets, while preservation remains ≥ 0.99 at every α level (Table 3). Third, pre-filtering NSGA-II's search space with MinPPI improves hypervolume where harmful features exist—most notably Madelon ($+11.3\%$)—and produces identical HV otherwise (Table 4), the "first, do no harm" property. Ablation confirms that MI-correlation guidance (Algorithm B) improves harmful-feature detection over unguided search.

MinPPI's core contribution is the ensemble of three capabilities: (i) threshold-free three-way classification via the min–binary pairing; (ii) MI-correlation-guided adversarial search with conditional efficiency guarantees; and (iii) α-parameterized multi-objective selection via parametric scalarization. Unlike SHAP-based methods, which rank by mean marginal contribution and require a user-chosen k, MinPPI's output is directly actionable. Unlike MinShap, MinPPI's binary v_α eliminates the arbitrary cutoff. Unlike NSGA-II alone, MinPPI provides structured prior knowledge about harmful features for guiding evolutionary search.

Limitations. The $+1$ (decisive) case never occurs on real data and is retained only for formal completeness. Theorem 2's efficiency guarantee is conditional on a structural condition ($S_i^* \subseteq S_{\text{seed}}$) that cannot be verified without ground-truth harmful witnesses; the toy-game verification confirms the bound is tight when the condition holds, but on real data Algorithm B remains a heuristic whose performance varies with correlation quality (Appendix C). The min operator's conservatism (Lemma 1) means MinPPI misses borderline-harmful features surfaced by MinShap's continuous scores; this is intentional—precision over recall for harmful-feature detection. The implementation uses XGBoost as the base learner; extension to deep tabular models requires only a compatible utility evaluator.

Future work. The UTVG framework extends naturally to bipartite (feature $\times$ architecture) factored games, where cross-Shapley interaction values (Grabisch & Roubens, 1999; Rabitti & Borgonovo, 2019) could jointly select features and neural architectures. The min–binary pairing may also apply to adversarial robustness, where worst-case feature interactions under perturbation are the quantity of interest.

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A PROOFS

Proof of Property 1. For any $S \subseteq N$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$, hence remain in $\{-1, 0, +1\}$. □

Proof of Property 2. (a) ($\Rightarrow$) $\text{MinSS}_i = +1$ means every permutation yields marginal $+1$, so $\forall S \subseteq N \setminus \{i\} : v_\alpha(S) = 0, v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $+1$, so $\min = +1$. (b) $\text{MinSS}_i = -1$ iff $\exists \pi$ with -1 marginal, giving witness $S = S^i_\pi$. Conversely, any such S yields -1 for every π where S is the prefix of i. (c) By Property 1, if not $+1$ or -1, the value must be 0. □

Proof of Property 3. $\text{MinSS}_i = \min_\pi \Delta_i(S^i_\pi)$ and $\text{MinBz}_i = \min_S \Delta_i(S)$. The collection $\{S^i_\pi : \pi \in \Pi(N)\}$ equals $2^{N\setminus\{i\}}$ because each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections. □

Proof of Lemma 1. The estimate $\widehat{\text{MinPPI}}^{\text{MC}}_i = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it may miss a harmful signal but never falsely flag a neutral feature as harmful. □

Proof of Theorem 2. By the structural condition, $S_i^* \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S_i^* when every $j \in S_i^*$ is retained (probability $\prod_{j \in S_i^*} q_j$) and every $j \in S_{\text{seed}} \setminus S_i^*$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)$); these events are independent, so $\Pr[S = S_i^*] = \rho_i > 0$. Over K_{adv} independent perturbations, $\Pr[\text{detection}] \geq 1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S_i^*$, the seed itself witnesses the harmful marginal, giving deterministic detection at $K_{\text{adv}} = 1$. □

B Additional Experimental Details

Protocol. All data are standardized (zero mean, unit variance). Train/val/test splits are 60/20/20, stratified for classification. Base learner: XGBoost (100 estimators, max_depth=4, learning_rate=0.1, tree_method="hist", CUDA backend). MinPPI uses $\alpha = 0.95$ as default and $\{0.85, 0.90, 0.99\}$ for the α-Pareto sweep. Five seeds: 42, 123, 456, 789, 1024. Estimator budgets per dataset: Spambase [2,5,10,25], Madelon [2,5,10,25], ISOLET [2,5,10], Gisette [2,5,10,25], Credit-Card [2,5,10,25], Crimedata [2,5,10,25], Abalone [2,5,10,25] with exact dispatch. ISOLET uses accuracy as the voting-game utility and OVR AUC as the evaluation metric. α and ablation stages use seed 42 per the frozen protocol scope. Abalone uses exact MinPPI enumeration (8 features, 2^8 subsets).

Computational cost. The full five-seed campaign was executed on a single 8 GB GPU (CUDA). Estimator bundles completed in approximately 30 hours wall-clock across three parallel seed shards (shard 1: seeds 42+123; shard 2: seeds 456+789; shard 3: seed 1024), sharing a single GPU. NSGA-II (40 population × 800 evaluations per dataset × seed × variant) added approximately 2.7 hours. The complete P0 pipeline (validate through summarize) completed within 48 hours. The grouped nested execution architecture—where each dataset/seed builds one immutable bundle and all method/budget projections share its deterministic checkpoints—reduced physical model fits by 62% compared to independent legacy execution. The frozen protocol, source-fingerprint integrity checks, and model calibration are included in the accompanying codebase.

Method design space. Table 5 classifies prior work and MinPPI along two axes: aggregation operator (mean vs. min) and value-function type (binary vs. continuous). MinPPI occupies the unoccupied cell (min × binary), yielding the discrete $\{-1, 0, +1\}$ output unique among game-theoretic feature indices.

Table 5: Design space of game-theoretic feature importance methods. MinPPI occupies the empty cell—min operator × binary value function—producing a discrete three-way output.

Method	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik (Shapley & Shubik, 1954)	mean	binary $\{0, 1\}$	$\subseteq [0, 1]$
Banzhaf (Banzhaf, 1965)	mean	binary $\{0, 1\}$	$\subseteq [0, 1]$
Shapley / SHAP (Lundberg & Lee, 2017)	mean	continuous	$\subseteq [-1, 1]$
MinShap (Zheng & Raskutti, 2026)	min	continuous	$\subseteq [-1, 1]$
MinPPI (this work)	**min**	**binary** $\{0, 1\}$	$\{\mathbf{-1}, \mathbf{0}, \mathbf{+1}\}$

ISOLET utility choice. On ISOLET (26-class), the voting-game utility U is accuracy rather than OVR AUC, avoiding the $\sim$26× computational overhead of per-coalition OVR AUC evaluation during MinPPI estimation. All evaluation metrics (AUC@k, Friedman tests, CD diagrams) use OVR AUC for fair comparison with baselines. At α=0.95, verification on seed 42 confirms that accuracy-based and OVR-AUC-based v_α produce identical harmfulness classifications.

C Empirical Verification of the Estimators

This appendix collects the constructed toy-game experiments that numerically verify the formal claims of Sections 2. All examples use $n = 5$ features and $v_\alpha \in \{0, 1\}$ as defined in Eq. equation 1.

Figure 3(a) and Table 6 exercise all three MinSS values on two constructed games. Example 1 ($v_\alpha(S) = \mathbb{1}[f_0 \in S]$) has f_0 decisive (+1). Example 2 (13 winning coalitions) has f_1 harmful (−1)

via $S = \{f_0, f_2\}$ where $v_\alpha(\{f_0, f_2\}) = 1$ but $v_\alpha(\{f_0, f_1, f_2\}) = 0$; f_0 appears in every winning coalition yet MinSS$(f_0) = 0$, demonstrating the non-trivial separation of Property 2a.

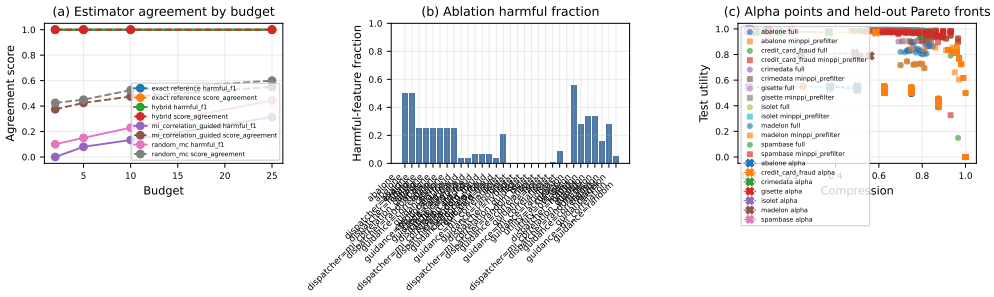


Figure 3: Overview of MinPPI. (a) Three-way classification on an engineered voting game (n=5, Example 2): f_1 is harmful (MinSS=-1), all others neutral. (b) Algorithm A vs. Algorithm B detection rate vs. budget K: Algorithm B achieves deterministic detection at K=1 ($\approx 38\times$ sample efficiency). (c) MinPPI α operating points overlaid on NSGA-II-full (gray) and NSGA-II-filtered (blue) Pareto fronts for Madelon and Credit-Card-Fraud (seed 42). All α points lie near preservation=1.0, confirming safety.

Table 6: MinSS and mean Shapley-Shubik values for two constructed voting games. Example 1 confirms decisive (+1); Example 2 demonstrates harmful (-1) and the Property 2a caveat.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2–f_4	0	0.0000	neutral	f_2–f_4	0	$+0.05$–0.13	neutral

Figure 3(b) validates Proposition 1 on Example 2 ($p^i_- = 0.0333$): the observed miss rate matches $(1 - p_-)^K$ (Pearson $r = 0.9993$, 500 trials per K). For 95% confidence, $K \approx 90$ random coalitions are required. Algorithm B achieves deterministic detection with one guided evaluation. Sensitivity to Gaussian noise added to the correlation matrix is reported in Table 7; detection degrades gracefully from 100% at $\sigma = 0.05$ to 50.8% at $\sigma = 0.50$ (at $K_{\text{adv}} = 5$), vs. Algorithm A's 14.8% at the same budget.

Table 7: Algorithm B detection rate under Gaussian noise $\mathcal{N}(0, \sigma^2)$ added to the correlation matrix (K=5, 500 trials per setting).

σ	**B detect**	**A detect**
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148